A sharp universal product threshold for Chambolle–Pock period-three orbits

Abstract

For the literal primal-first Chambolle–Pock map with any nonnegative extrapolation parameter, we determine the smallest stepsize product that permits a nonconstant period-three orbit. Necessity holds for arbitrary proper closed convex functions, arbitrary finite-dimensional real Hilbert spaces, and arbitrary linear couplings, without a saddle-point assumption. A centered-triple shift identity, two cyclic subgradient inequalities, and a Gram bound reduce the proof to an ellipse support calculation. Planar indicator functions attain the bound with an existing saddle point and an explicit paired-state initialization. This strengthens the restricted ordered-triangle optimality result of report 18. It is a universal period-three threshold, not a universal convergence threshold. No novelty claim is made.

Keywords: Chambolle–Pock iteration; period-three threshold; cyclic subgradient inequalities; convex interpolation; sharp product bound

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1 Setting and sharp threshold

Let X, Y be finite-dimensional real Hilbert spaces, with their given inner products, and let $L : X \to Y$ be a bounded linear map. Let $f : X \to (-\infty, +\infty]$ and $h : Y \to (-\infty, +\infty]$ be proper closed convex functions. We write $h = g^*$; conversely any such h is the conjugate of $g = h^*$. For $a > 0$,

$$\operatorname{prox}_{af}(z) = \arg\min_x \left\{ f(x) + \frac{\|x - z\|^2}{2a} \right\}.$$

Fix $t \geq 0$ and $\tau, \sigma > 0$. The literal paired-state map is

$$\begin{aligned} x_{k+1} &= \operatorname{prox}_{\tau f}(x_k - \tau L^* y_k), \\ y_{k+1} &= \operatorname{prox}_{\sigma h}\big(y_k + \sigma L((1+t)x_{k+1} - t x_k)\big). \end{aligned} \tag{1}$$

The newly computed primal point is used in the dual update. There is no outer relaxation or additional state. We use the primal–dual framework of Chambolle and Pock [1], specifying the exact order above.

Put

$$W = 1 + 3t + 3t^2, \qquad s = \sqrt{W}, \qquad \Delta = 2s - 1 - 3t, \qquad r_t = \frac{3}{\Delta}. \tag{2}$$

The identity $4W - (1+3t)^2 = 3(1+t)^2 > 0$ proves $\Delta > 0$ for $t \geq 0$.

Theorem 1. *If* (1) *has a nonconstant orbit satisfying* $(x_{k+3}, y_{k+3}) = (x_k, y_k)$ *for all* $k \geq 0$, *then*

$$\tau\sigma\|L\|^2 \geq r_t = \frac{2\sqrt{1+3t+3t^2} + 1 + 3t}{(1+t)^2}. \tag{3}$$

This implication does not require the existence of a saddle point.

For each $t \geq 0$, *equality is attained by indicator functions of two nondegenerate triangles in ordinary Euclidean* $\mathbb{R}^2$, *with* $L = I$, $\sigma = 1$, $\tau = r_t$, *and a saddle point at* $(0, 0)$. *An explicit initialization produces an orbit of least period three.*

Consequently r_t *is the minimum product admitting a nonconstant period-three orbit over the entire stated convex class. The minimum is the same if saddle existence is imposed.*

A nonconstant orbit with period dividing three necessarily has least period three: the only positive divisors of three are one and three. All the necessity arguments below apply to arbitrary three states of such an orbit; no regular-triangle or planar hypothesis is imposed.

2 Cyclic inequalities and centered triples

Index the three successive states by i modulo three. Each state is an output of a proximal map, including phase zero because of periodicity. Thus $f(x_i)$ and $h(y_i)$ are finite. Proximal optimality gives

$$\begin{aligned} a_i &= (x_{i-1} - x_i)/\tau - L^* y_{i-1} \in \partial f(x_i), \\ b_i &= (y_{i-1} - y_i)/\sigma + L((1+t)x_i - t x_{i-1}) \in \partial h(y_i). \end{aligned} \tag{4}$$

The inequalities $f(x_{i+1}) \geq f(x_i) + \langle a_i, x_{i+1} - x_i \rangle$ telescope, and the same holds for h. Their precise orientation is

$$\sum_i \langle a_i, x_{i+1} - x_i \rangle \leq 0, \qquad \sum_i \langle b_i, y_{i+1} - y_i \rangle \leq 0. \tag{5}$$

The three function values need not be equal.

Define the means $\bar{x} = (x_0 + x_1 + x_2)/3$, $\bar{y} = (y_0 + y_1 + y_2)/3$ and centered triples

$$U = (x_i - \bar{x})_{i=0}^2 \in X^3, \qquad V = (y_i - \bar{y})_{i=0}^2 \in Y^3.$$

Use the sum inner product on each direct sum. On triples, let the backward shift be $(Sz)_i = z_{i-1}$, so $S^* = S^2$ is the forward shift. On the centered subspaces,

$$I + S + S^* = 0, \qquad J = \frac{S - S^*}{\sqrt{3}}, \qquad J^* = -J, \quad J^2 = -I, \quad S = -\frac{1}{2}I + \frac{\sqrt{3}}{2}J. \tag{6}$$

Indeed $S^3 = I$ and the centered sum vanishes. Then $(S - S^*)^2 = S + S^* - 2I = -3I$, proving the assertion about J^2. In particular $J^*J = I$ on centered triples.

Let $\mathcal{L} : X^3 \to Y^3$ be the diagonal lift of L. Its norm is $\ell = \|L\|$, and it preserves centered triples. It commutes with shifts and satisfies $\mathcal{L}J_X = J_Y\mathcal{L}$. Subscripts on J and S will be omitted when their spaces are clear. Thus $\mathcal{L}U$ and $\mathcal{L}JU = J\mathcal{L}U$ are orthogonal and have equal norm. Explicitly,

$$\langle \mathcal{L}U, J\mathcal{L}U \rangle = 0, \qquad \|J\mathcal{L}U\|^2 = \|\mathcal{L}U\|^2.$$

This uses skew-adjointness and orthogonality of J, not an isometry assumption on L.

Set

$$A = \|U\|^2, \quad B = \|V\|^2, \quad c = \langle V, \mathcal{L}U \rangle, \quad d = \langle V, \mathcal{L}JU \rangle. \tag{7}$$

Centering introduces only constant triples in the linear terms of (5); their inner products with difference triples vanish. Hence those inequalities are exactly

$$\tau^{-1}\langle (S - I)U, (S^* - I)U \rangle - \langle \mathcal{L}^*SV, (S^* - I)U \rangle \le 0, \tag{8}$$

$$\sigma^{-1}\langle (S - I)V, (S^* - I)V \rangle + \langle \mathcal{L}((1 + t)I - tS)U, (S^* - I)V \rangle \le 0. \tag{9}$$

3 Exact shift algebra and the Gram bound

For every centered triple Z, the symmetric part of $(S^* - I)^2$ is $(3/2)I$. Consequently

$$\langle (S - I)Z, (S^* - I)Z \rangle = \frac{3}{2}\|Z\|^2.$$

For the cross term in (8), commuting the lift with the shift and moving the shift across the inner product gives

$$\langle \mathcal{L}^* SV, (S^* - I)U \rangle = \langle V, \mathcal{L}S^*(S^* - I)U \rangle = \langle V, \mathcal{L}(S - S^*)U \rangle = \sqrt{3}d.$$

For the remaining term, moving the forward difference across gives

$$\begin{aligned} &\langle \mathcal{L}((1+t)I - tS)U, (S^* - I)V \rangle \\ &\quad = \langle V, \mathcal{L}(S - I)((1+t)I - tS)U \rangle \\ &\quad = -\frac{3}{2}(1+t)c + \frac{\sqrt{3}}{2}(1+3t)d, \end{aligned}$$

since, on centered triples,

$$(S - I)((1+t)I - tS) = (1+3t)S - I = -\frac{3}{2}(1+t)I + \frac{\sqrt{3}}{2}(1+3t)J.$$

Substituting these identities into (8)–(9) yields

$$d \geq \frac{\sqrt{3}A}{2\tau}, \qquad \frac{B}{\sigma} - (1+t)c + \frac{1+3t}{\sqrt{3}}d \leq 0. \tag{10}$$

If $A = 0$, then $c = d = 0$ and the second inequality forces $B = 0$. Thus a nonconstant three-cycle has $A > 0$. If $\ell = 0$, then $d = 0$, contradicting the first inequality. Hence $\ell > 0$ as well. These observations cover both the primal-constant and zero-coupling degeneracies.

Bessel's inequality for the orthogonal equal-length vectors $\mathcal{L}U$ and $\mathcal{L}JU$ now gives

$$c^2 + d^2 \leq \|\mathcal{L}U\|^2 B \leq \ell^2 AB. \tag{11}$$

For completeness, when their common norm is positive, divide both vectors by that norm and apply the orthogonal-projection inequality to V. When it is zero, $c = d = 0$ and (11) is immediate. In a nonconstant orbit (10) actually excludes this latter case. No division by a potentially vanishing vector norm is required.

4 Ellipse support and universal necessity

Combining (10) and (11) gives

$$c^2 + d^2 \le \sigma\ell^2 A \left((1+t)c - \frac{1+3t}{\sqrt{3}} d \right).$$

The positive factors established above allow the definitions

$$u = \frac{c}{\sigma\ell^2 A}, \qquad w = \frac{\sqrt{3}d}{\sigma\ell^2 A}, \qquad r = \tau\sigma\ell^2 > 0.$$

The necessary scalar conditions become

$$3u^2 + w^2 - 3(1+t)u + (1+3t)w \le 0, \qquad rw \ge \frac{3}{2}. \tag{12}$$

Completing the square in the first condition gives precisely

$$3\left(u - \frac{1+t}{2}\right)^2 + \left(w + \frac{1+3t}{2}\right)^2 \le W. \tag{13}$$

Therefore

$$w \le \sqrt{W} - \frac{1+3t}{2} = \frac{\Delta}{2}.$$

The second condition in (12) implies $w > 0$, so

$$r \ge \frac{3}{2w} \ge \frac{3}{\Delta} = r_t.$$

Rationalizing $3/\Delta$ using $4W - (1+3t)^2 = 3(1+t)^2$ yields the second expression in (3). This proves the necessary part of Theorem 1, without any assumption of a saddle point.

The scalars u, w in this reduction describe correlations of arbitrary centered triples. They are not coordinates of assumed planar vertices. Their appearance in the ellipse is a consequence of convexity and the three-phase shift algebra. The geometric construction next realizes its maximal w and simultaneously satisfies the remaining projection tests.

5 Sharp attainment in the Euclidean plane

Take $X = Y = \mathbb{R}^2$ with its ordinary inner product and set

$$R = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad u = \frac{1+t}{2}, \quad w = \frac{\Delta}{2}, \quad r = \frac{3}{\Delta}.$$

Let $c_i = R^i(1,0)$, $d_i = R^i(u, -w/\sqrt{3})$ for $i = 0, 1, 2$.

Define

$$C = \operatorname{conv}\{c_0, c_1, c_2\}, \qquad D = \operatorname{conv}\{d_0, d_1, d_2\},$$

and use $f = \iota_C$, $h = \iota_D$, $L = I$, $\sigma = 1$, $\tau = r$. Here ι_E equals zero on E and $+\infty$ elsewhere. Both triangles are nondegenerate: the initial primal vertex is nonzero and $u > 0$ makes the initial dual vertex nonzero. Both functions are proper closed convex, and their proximal maps are the Euclidean projections P_C and P_D, independently of the positive proximal parameter.

For a closed convex set K, a point $z \in K$ satisfies $z = P_K(q)$ exactly when

$$\langle q - z, z' - z \rangle \le 0 \qquad (z' \in K). \tag{14}$$

Sufficiency follows by expanding squared distances; necessity follows by testing $z+\varepsilon(z'-z)$ and letting $\varepsilon \downarrow 0$. Squared distance is strictly convex, so the projection is unique. For a convex hull it suffices to test the vertices, by affinity of the left side.

For the transition $(c_2, d_2) \mapsto (c_0, d_0)$ the exact residuals are

$$p = c_2 - r d_2 - c_0, \qquad q = d_2 + (1+t)c_0 - t c_2 - d_0.$$

Direct multiplication by R gives the four tests, ordered by target vertices $1, 2$:

$$\langle p, c_1 - c_0 \rangle = \frac{3}{2} - rw, \tag{15}$$

$$\langle p, c_2 - c_0 \rangle = \frac{6 - r(3u + w)}{2}, \tag{16}$$

$$\langle q, d_1 - d_0 \rangle = \frac{3u^2 + w^2 - 3(1+t)u + (1+3t)w}{2}, \tag{17}$$

$$\langle q, d_2 - d_0 \rangle = 3u^2 + w^2 - \left(3t + \frac{3}{2}\right) u - \frac{w}{2}. \tag{18}$$

The choices above give zero in (15) and (17). The remaining expressions are

$$6 - r(3u + w) = \frac{9(s - 2t - 1)}{\Delta}, \tag{19}$$

$$3u^2 + w^2 - \left(3t + \frac{3}{2}\right) u - \frac{w}{2} = \frac{3}{2} s(s - 2t - 1). \tag{20}$$

Since $(2t+1)^2 - s^2 = t(t+1) \ge 0$ and $s, 2t+1 > 0$, both are nonpositive. Thus (14) proves the desired transition.

Rotation preserves the inner products and both triangles, so the same tests prove $(c_{i-1}, d_{i-1}) \mapsto (c_i, d_i)$ for every phase. At $t = 0$ all four tests are equalities; this causes no ambiguity because the projections remain unique.

Initialize the literal map at $(x_0, y_0) = (c_2, d_2)$. Induction now gives, for every $k \geq 0$,

$$\begin{aligned} (x_{3k}, y_{3k}) &= (c_2, d_2), \\ (x_{3k+1}, y_{3k+1}) &= (c_0, d_0), \\ (x_{3k+2}, y_{3k+2}) &= (c_1, d_1). \end{aligned} \tag{21}$$

No prehistory or extrapolated-state initialization is needed. The primal vertices are distinct, so the least period is exactly three.

Because $I + R + R^2 = 0$, both vertex sums vanish and $0 \in C \cap D$. In particular

$$-L^*0 = 0 \in \partial f(0) = N_C(0), \qquad L0 = 0 \in \partial h(0) = N_D(0).$$

These are the saddle/KKT conditions. More explicitly, $g = h^*$ is the support function $g(z) = \max_i \langle z, d_i \rangle$, and the Lagrangian $\mathcal{F}(x, y) = f(x) + \langle Lx, y \rangle - h(y)$ satisfies

$$\mathcal{F}(0, y) \leq \mathcal{F}(0, 0) = 0 \leq \mathcal{F}(x, 0) \qquad (x \in X,\ y \in Y).$$

Thus $(0, 0)$ is a saddle pair. Finally $\|I\| = 1$ in the actual Euclidean norm, and $\tau\sigma\|L\|^2 = r_t$. This proves attainment and completes Theorem 1.

6 Consequences and exact scope

Report 18, *Exact planar Chambolle–Pock three-cycles and an optimal ordered triangle family*, established optimality within a prescribed ordered equilateral-triangle class. Theorem 1 strictly broadens that optimality statement: its lower bound applies to every nonconstant period-three orbit in the entire convex class specified here, in arbitrary finite dimensions and with arbitrary couplings. The planar construction already attains this universal lower bound.

Nevertheless, a sharp minimum for period three does not determine the universal convergence threshold. Below r_t it excludes period-three orbits only. It does not exclude other periods, unbounded trajectories, or bounded aperiodic failure of convergence, and it does not establish convergence from all initial states.

At the endpoint $t = 0$ the formula gives $r_0 = 3$. At the example parameter $t = 1/8$ it gives

$$r_{1/8} = \frac{88 + 16\sqrt{91}}{81} \approx 2.970744, \qquad d_0 = \left(\frac{9}{16}, \frac{11 - 2\sqrt{91}}{16\sqrt{3}} \right).$$

These are now universally minimal period-three products, rather than merely minimal products for the ordered-triangle ansatz. The occurrence of a nonconvergent orbit at equality shows that a full-iterate convergence guarantee covering every problem and every initialization cannot include this product. It does not settle any lower convergence boundary.

The attaining orbits are bounded, and their ordinary Cesàro averages converge to $(0, 0)$: every block of three joint states sums to zero, while the finite remainder divided by the number of iterates tends to zero. There is no claim here of an attracting cycle, robustness under perturbed initialization, a classification of equality cases, or a result for a different update order or relaxation.

6.1 Reproducibility

The portable script `verify_exact.py` requires Python 3 and SymPy. Run `python3 verify_exact.py`. It checks the centered-shift identities with exact matrices, the primal and dual residual products directly from the planar update, the ellipse completion, and the sharpness and positivity identities. The argument for arbitrary convex functions is the subgradient and Hilbert-space proof above; symbolic checks supplement rather than replace it. Signs for all $t \geq 0$ follow from explicit square identities, not numerical sampling.

Run `bash build.sh` with a standard pdfLaTeX installation to reproduce this manuscript. The source package includes the manuscript, the reference-style file, the build script, and the exact verifier. It contains no numerical optimization output needed for the proof.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.